PHYSICS GROUP PROJECT REPORT

Mathematical Derivation and Numerical Verification of the EAPEF for Motion on a Surface of Revolution

Timberlake & Mbenoun, Am. J. Phys. ***94****, 16 (2026)*

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June 17, 2026

Abstract

We derive the effective analogous potential energy function (EAPEF) for a point mass on a frictionless surface of revolution in uniform gravity [1], and check the result by explicit Euler integration ($\Delta t = 0.005\,\mathrm{s}$, $m = g = 1$). The report gives the reduced Lagrangian, conservation of L_z, the EAPEF, the radial equation of motion, circular-orbit conditions, and the stability indicator $S(r) = 3z' + rz''$.

For the cone $z = r$ and the paraboloid $z = r^2$, turning radii agree with $U_{\mathrm{eq}} = 0$ to within 0.04%, energy-error scaling with Δt is consistent with first-order Euler integration, and the horizontal projection precesses ($\langle \Delta\phi \rangle/(2\pi) = 0.80$ and 1.86 per radial cycle). For the Gaussian profile $z = e^{-r^2}$, the EAPEF has a single zero and the particle escapes outward, in agreement with the analytic prediction.

Contents

1 Introduction

Timberlake and Mbenoun [1] introduce an effective analogous potential $U_{\rm eq}(r)$ for analyzing motion on a surface of revolution $z = z(r)$ in a uniform gravitational field. This report has two parts: a step-by-step analytic derivation, and numerical integration of the resulting equations.

We study three surfaces:

- **Cone** ($z = r$) and **paraboloid** ($z = r^2$): each has two zeros of $U_{\rm eq}(r)$, so the particle oscillates between inner and outer turning radii. We compare predicted and simulated turning points, energy drift, and azimuthal motion.
- **Gaussian bell** ($z = e^{-r^2}$): the surface slopes downward for $r > 0$ and yields only one EAPEF zero. We use this case to test whether the framework correctly predicts outward escape.

Numerical results in Secs. 5–7 were obtained with the explicit Euler scheme in Sec. 5.1 ($\Delta t = 0.005\,\mathrm{s}$, $m = g = 1$).

2 Physical Setup and Coordinates

A point particle of mass m moves without friction on a surface of revolution defined by the holonomic constraint

$$z = z(r), \qquad r \geq 0, \tag{1}$$

where r and ϕ are cylindrical coordinates and z is the vertical height. Gravity acts uniformly with acceleration g in the $-z$ direction, giving potential energy

$$V(r) = mg\,z(r). \tag{2}$$

No dissipative forces are present. The configuration space is two-dimensional: (r, ϕ), with z slaved to r through (1).

Table 1: Symbols used throughout. Numerical runs use $m = 1\,\mathrm{kg}$, $g = 1\,\mathrm{m/s^2}$.

Symbol	Meaning	Units
L_z	Conserved azimuthal angular momentum	$\mathrm{kg\,m^2/s}$
E	Total mechanical energy	J
$U_{\rm eq}(r)$	Effective analogous potential (EAPEF)	J
$S(r)$	Stability indicator $3z' + rz''$	dimensionless

3 Complete Mathematical Derivation

This section presents the full derivation chain from cylindrical coordinates to the radial equation and stability criterion.

3.1 Kinetic Energy in Cylindrical Coordinates

Setup. Express the kinetic energy T in terms of (r, ϕ, z) and their time derivatives.

The position vector in cylindrical coordinates is

$$\mathbf{r} = r\cos\phi\,\hat{\mathbf{x}} + r\sin\phi\,\hat{\mathbf{y}} + z\,\hat{\mathbf{z}}. \tag{3}$$

Differentiating each component,

$$\dot{x} = \dot{r}\cos\phi - r\dot{\phi}\sin\phi, \qquad \dot{y} = \dot{r}\sin\phi + r\dot{\phi}\cos\phi, \qquad \dot{z} = \dot{z}. \tag{4}$$

The squared speed is the sum of squared components:

$$\dot{\mathbf{r}}^{\,2} = (\dot{r}\cos\phi - r\dot{\phi}\sin\phi)^2 + (\dot{r}\sin\phi + r\dot{\phi}\cos\phi)^2 + \dot{z}^2. \tag{5}$$

Expanding the first square,

$$(\dot{r}\cos\phi - r\dot{\phi}\sin\phi)^2 = \dot{r}^2\cos^2\phi - 2\dot{r}r\dot{\phi}\cos\phi\sin\phi + r^2\dot{\phi}^2\sin^2\phi. \tag{6}$$

Expanding the second square,

$$(\dot{r}\sin\phi + r\dot{\phi}\cos\phi)^2 = \dot{r}^2\sin^2\phi + 2\dot{r}r\dot{\phi}\cos\phi\sin\phi + r^2\dot{\phi}^2\cos^2\phi. \tag{7}$$

Adding these two expressions, the cross terms cancel:

$$\dot{r}^2(\cos^2\phi + \sin^2\phi) + r^2\dot{\phi}^2(\sin^2\phi + \cos^2\phi) = \dot{r}^2 + r^2\dot{\phi}^2. \tag{8}$$

Therefore

$$\dot{\mathbf{r}}^{\,2} = \dot{r}^2 + r^2\dot{\phi}^2 + \dot{z}^2, \tag{9}$$

and the kinetic energy is

$$T = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\phi}^2 + \dot{z}^2\right). \tag{10}$$

Physical Interpretation

Equation (10) is the starting point before imposing the surface constraint. The azimuthal term $r^2\dot{\phi}^2$ will produce the centrifugal barrier; the vertical term $\dot{z}^2$ will be eliminated using $z = z(r)$.

3.2 Constraint Substitution

Setup. Eliminate $\dot{z}$ in favor of $\dot{r}$ using the surface constraint (1).

Since z depends only on r, the chain rule gives

$$\dot{z} = \frac{dz}{dt} = \frac{dz}{dr}\frac{dr}{dt} = z'(r)\,\dot{r}. \tag{11}$$

This step is necessary because the particle cannot move independently in z; vertical motion is locked to radial motion along the slope of the surface.

Squaring (11),

$$\dot{z}^2 = [z'(r)]^2\,\dot{r}^2. \tag{12}$$

3.3 Reduced Kinetic Energy

Setup. Obtain T as a function of $(r, \dot{r}, \dot{\phi})$ only.

Substitute (12) into (10):

$$T = \frac{1}{2}m\left[\dot{r}^2 + r^2\dot{\phi}^2 + [z'(r)]^2\,\dot{r}^2\right]$$
$$= \frac{1}{2}m\left[\dot{r}^2\left(1 + [z'(r)]^2\right) + r^2\dot{\phi}^2\right]. \tag{13}$$

Physical meaning. The factor $1 + [z'(r)]^2$ is a *metric correction*: moving a distance $\dot{r}\,dt$ in the radial direction also changes z by $z'\dot{r}\,dt$, so the total path element includes both horizontal and vertical contributions. Effective inertia in the radial direction exceeds m when the surface is steep.

3.4 Lagrangian Construction

Setup. Form $\mathcal{L} = T - V$ for use in the Euler–Lagrange equations.

With $V = mgz(r)$ from (2) and T from (13),

$$\mathcal{L}(r, \dot{r}, \phi, \dot{\phi}) = \frac{1}{2}m\left\{\dot{r}^2\left(1 + [z'(r)]^2\right) + r^2\dot{\phi}^2\right\} - mg\,z(r). \tag{14}$$

Note that $\mathcal{L}$ depends on ϕ only through $\dot{\phi}$, not explicitly on ϕ itself.

3.5 Conservation of Azimuthal Angular Momentum

Setup. Show that $L_z = mr^2\dot{\phi}$ is a constant of motion.

Because $\partial\mathcal{L}/\partial\phi = 0$, the coordinate ϕ is *cyclic*. The conjugate momentum is

$$p_\phi = \frac{\partial\mathcal{L}}{\partial\dot{\phi}} = \frac{\partial}{\partial\dot{\phi}}\left(\frac{1}{2}mr^2\dot{\phi}^2\right) = mr^2\dot{\phi}. \tag{15}$$

The Euler–Lagrange equation for ϕ is

$$\frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{\phi}}\right) - \frac{\partial\mathcal{L}}{\partial\phi} = 0. \tag{16}$$

Since $\partial\mathcal{L}/\partial\phi = 0$,

$$\frac{d}{dt}(mr^2\dot{\phi}) = 0 \quad\Longrightarrow\quad mr^2\dot{\phi} = \text{constant}. \tag{17}$$

We define the conserved azimuthal angular momentum

$$L_z \equiv mr^2\dot{\phi}, \tag{18}$$

so that

$$\dot{\phi} = \frac{L_z}{mr^2}. \tag{19}$$

Physical meaning. Rotational symmetry about the z axis implies zero azimuthal torque; therefore L_z is conserved for all surface profiles $z(r)$.

3.6 Total Energy and the Effective Analogous Potential

Setup. Reduce the dynamics to a one-dimensional radial problem by eliminating $\dot{\phi}$ and constructing $U_{\rm eq}(r)$.

Substitute (19) into (13):

$$
\begin{aligned}
T &= \frac{1}{2}m\dot{r}^2\left(1+[z'(r)]^2\right)+\frac{1}{2}mr^2\left(\frac{L_z}{mr^2}\right)^2 \\
&= \frac{1}{2}m\dot{r}^2\left(1+[z'(r)]^2\right)+\frac{L_z^2}{2mr^2}.
\end{aligned}
\tag{20}
$$

Adding $V = mgz(r)$, the total energy is

$$
E = \frac{1}{2}m\dot{r}^2\left(1+[z'(r)]^2\right)+\frac{L_z^2}{2mr^2}+mg\,z(r).
\tag{21}
$$

Isolating $\dot{r}^2$. This step is necessary to interpret radial motion as motion in a one-dimensional potential. Rearranging (21):

$$
\frac{1}{2}m\dot{r}^2\left(1+[z'(r)]^2\right) = E-\frac{L_z^2}{2mr^2}-mg\,z(r).
\tag{22}
$$

Divide both sides by $\frac{1}{2}m(1+[z']^2)$:

$$
\dot{r}^2 = \frac{2\left[E-\dfrac{L_z^2}{2mr^2}-mg\,z(r)\right]}{m\left(1+[z'(r)]^2\right)}.
\tag{23}
$$

Expand the numerator of (23) by multiplying through by mr^2:

$$
\begin{aligned}
2\left[E-\frac{L_z^2}{2mr^2}-mg\,z(r)\right] &= \frac{2Emr^2-L_z^2-2mr^2mgz(r)}{mr^2} \\
&= \frac{2mr^2(E-mgz)-L_z^2}{mr^2}.
\end{aligned}
\tag{24}
$$

Note that $2mr^2(E-mgz)-L_z^2 = -\left[L_z^2-2mr^2(E-mgz)\right]$. Substituting into (23):

$$
\dot{r}^2 = \frac{2mr^2(E-mgz)-L_z^2}{m^2r^2\left(1+[z'(r)]^2\right)} = \frac{-\left[L_z^2-2mr^2\left(E-mg\,z(r)\right)\right]}{m^2r^2\left(1+[z'(r)]^2\right)}.
\tag{25}
$$

Following Ref. [1], define the EAPEF:

Effective Analogous Potential (EAPEF)

$$
U_{\rm eq}(r) = \frac{L_z^2-2mr^2\left(E-mg\,z(r)\right)}{2mr^2\left(1+[z'(r)]^2\right)}.
\tag{26}
$$

Then

$$
\dot{r}^2 = -\frac{2U_{\rm eq}(r)}{m},
\tag{27}
$$

so that classically allowed regions satisfy $U_{\rm eq}(r) \le 0$, and turning points occur at $U_{\rm eq}(r) = 0$.

Physical Interpretation

Equation (26) is needed because the particle's radial kinetic energy is not $\frac{1}{2}m\dot{r}^2$ but $\frac{1}{2}m\dot{r}^2(1+[z']^2)$: the surface geometry modifies the effective inertia. It plays the same *conceptual* role as the familiar effective potential $U_{\text{eff}} = L_z^2/(2mr^2) + mgz(r)$ in central-force problems. The centrifugal barrier $L_z^2/(2mr^2)$ and gravitational term $mgz(r)$ appear implicitly inside the numerator, but the denominator $1 + [z'(r)]^2$ is the new ingredient absent from standard central-force treatments.

Compared with U_{eff}, the EAPEF enforces the correct turning-point condition $\dot{r} = 0$ at $U_{\text{eq}} = 0$ while respecting the metric factor. On the cone at $r_0 = 1.2\,\text{m}$, $U_{\text{eq}}(r_0) = -0.125\,\text{J}$ and $U_{\text{eff}}(r_0) = -0.25\,\text{J}$ at the same (E_0, L_z) illustrate the geometric correction directly.

3.7 Radial Equation from the Euler–Lagrange Equation

Setup. Derive $\ddot{r}$ as a function of $(r, \dot{r}, L_z)$ for numerical integration.

For the non-cyclic coordinate r, the Euler–Lagrange equation is

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{r}}\right) - \frac{\partial \mathcal{L}}{\partial r} = 0. \tag{28}$$

Euler–Lagrange Derivation of $\ddot{r}$

Step 1: $\partial\mathcal{L}/\partial\dot{r}$. From (14),

$$\frac{\partial \mathcal{L}}{\partial \dot{r}} = m\dot{r}\left(1 + [z'(r)]^2\right). \tag{29}$$

Differentiating with respect to time (remembering that $z' = z'(r)$ depends on r):

$$\frac{d}{dt}\left(1 + [z'(r)]^2\right) = \frac{d}{dr}\left(1 + [z'(r)]^2\right)\frac{dr}{dt} = 2z'(r)z''(r)\,\dot{r}. \tag{30}$$

Therefore

$$\begin{aligned}\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{r}}\right) &= m\ddot{r}\left(1 + [z']^2\right) + m\dot{r}\left[2z'z''\dot{r}\right] \\ &= m\ddot{r}\left(1 + [z']^2\right) + 2m\dot{r}^2 z'z''. \end{aligned} \tag{31}$$

The second term arises because the effective inertia coefficient $1 + [z']^2$ changes as the particle moves along the surface.

Step 2: $\partial\mathcal{L}/\partial r$. Because $z(r)$, $z'(r)$, and $z''(r)$ all depend on r, differentiate (14) term by term (holding $\dot{r}$ and $\dot{\phi}$ fixed in the partial derivative):

$$\frac{\partial}{\partial r}\left[\frac{1}{2}m\dot{r}^2\left(1 + [z']^2\right)\right] = \frac{1}{2}m\dot{r}^2 \cdot \frac{d}{dr}\left(1 + [z']^2\right) = \frac{1}{2}m\dot{r}^2 \cdot 2z'z'' = m\dot{r}^2 z'z'', \tag{32}$$

$$\frac{\partial}{\partial r}\left[\frac{1}{2}mr^2\dot{\phi}^2\right] = \frac{L_z^2}{2m}\frac{\partial(r^{-2})}{\partial r} = \frac{L_z^2}{2m}(-2r^{-3}) = -\frac{L_z^2}{mr^3}, \tag{33}$$

$$\frac{\partial}{\partial r}\left[-mgz(r)\right] = -mgz'(r). \tag{34}$$

Summing,

$$\frac{\partial \mathcal{L}}{\partial r} = m\dot{r}^2 z' z'' - \frac{L_z^2}{mr^3} - mgz'(r). \tag{35}$$

The first term is a *geometric force* from the changing slope; the second is the centrifugal term written using L_z; the third is gravity projected along the surface.

Step 3: Assemble (28). Substituting (31) and (35) into (28):

$$\underbrace{m\ddot{r}(1+[z']^2) + 2m\dot{r}^2 z' z''}_{d(\partial\mathcal{L}/\partial\dot{r})/dt} - \underbrace{\left(m\dot{r}^2 z' z'' - \frac{L_z^2}{mr^3} - mgz'\right)}_{\partial\mathcal{L}/\partial r} = 0. \tag{36}$$

Collecting terms:

$$m\ddot{r}(1+[z']^2) + m\dot{r}^2 z' z'' + \frac{L_z^2}{mr^3} + mgz' = 0. \tag{37}$$

Divide by $m(1+[z']^2)$ (assumption: surface not vertical, so $1+[z']^2 \neq 0$):

$$\ddot{r} = \frac{\dfrac{L_z^2}{m^2r^3} - gz'(r) - z'(r)z''(r)\,\dot{r}^2}{1+[z'(r)]^2}. \tag{38}$$

Physical Interpretation

This radial ODE is the dynamical law integrated in our simulations. The term $L_z^2/(m^2r^3)$ is the azimuthal centripetal requirement: larger L_z or smaller r pushes the particle outward along the surface. The term $-gz'(r)$ is gravity resolved along the surface tangent—on an upward-sloping cone ($z'=1$), it acts to reduce $\ddot{r}$. The term $-z'z''\dot{r}^2$ is purely geometric: it vanishes for a cone ($z''=0$) but becomes significant on the paraboloid ($z''=2$), where large radial speed $\dot{r}$ couples to curvature and modifies the effective radial force. Unlike planar central-force problems, this velocity-dependent correction has no analogue in $U_{\text{eff}}(r)$ alone; it must be retained in the ODE even when $U_{\text{eq}}(r)$ correctly predicts turning points.

3.8 Circular Orbit Conditions

Setup. Find conditions for a circular orbit at constant radius $r = r_c$.

A circular orbit requires $\dot{r}=0$ and $\ddot{r}=0$ for all time. We apply each condition to a different equation.

Condition 1: $U_{\text{eq}}(r_c)=0$ with $\dot{r}=0$. From (27), $\dot{r}=0$ requires $U_{\text{eq}}(r_c)=0$:

$$L_z^2 - 2mr_c^2\,[E - mg\,z(r_c)] = 0 \quad\Longleftrightarrow\quad L_z^2 = 2mr_c^2\,[E - mg\,z(r_c)]\,. \tag{39}$$

Physically, the particle sits on the "energy surface" defined by the EAPEF at fixed r_c.

Condition 2: $\ddot{r}=0$ with $\dot{r}=0$. Setting $\dot{r}=0$ in (38) removes the velocity-dependent curvature term:

$$0 = \frac{L_z^2/(m^2r_c^3) - gz'(r_c)}{1+[z'(r_c)]^2} \quad\Longrightarrow\quad \frac{L_z^2}{m^2r_c^3} = gz'(r_c). \tag{40}$$

This is the radial force balance: azimuthal centripetal requirement equals the component of gravity along the surface. Equivalently,

$$L_z^2 = mgr_c^3 z'(r_c). \tag{41}$$

Condition 3: Consistency of energy and angular momentum. Equating the right-hand sides of (39) and (41):

$$2mr_c^2 \left[E - mg\, z(r_c)\right] = mgr_c^3 z'(r_c). \tag{42}$$

Divide by mr_c^2:

$$2\left[E - mg\, z(r_c)\right] = gr_c z'(r_c) \quad \Longrightarrow \quad E = mg\, z(r_c) + \frac{mg}{2}\, r_c z'(r_c). \tag{43}$$

Substituting (43) back into (41):

$$L_z^2 = mgr_c^3 z'(r_c) \quad \Longrightarrow \quad L_z = \sqrt{mgr_c^3 z'(r_c)}. \tag{44}$$

For L_z to be real with $r_c > 0$, we require $z'(r_c) > 0$ (surface height increasing with r).

Physical Interpretation

Equation (44) specifies the unique angular momentum for which a particle can remain at fixed radius: centrifugal push $L_z^2/(mr_c^3)$ exactly balances the downhill gravitational pull along the surface, $gz'(r_c)$. It is the surface-of-revolution analogue of the circular-orbit speed condition $v = \sqrt{gm/r}$ in horizontal circular motion, but modified by the slope $z'(r_c)$. When the simulated L_z exceeds $L_{z,c}$—as in the paraboloid run ($L_z = 5.0$ vs. $L_{z,c} = 2.036$)—the centrifugal barrier dominates and the orbit develops large radial excursions between the turning radii 1.2000 m and 2.9459 m.

3.9 Stability Criterion for Circular Orbits

Setup. Derive the condition under which a circular orbit at r_c is a stable equilibrium of the radial motion.

Stability requires $U'_{\text{eq}}(r_c) = 0$ and $U''_{\text{eq}}(r_c) \geq 0$. Write

$$U_{\text{eq}}(r) = \frac{f(r)}{g(r)}, \qquad f(r) = L_z^2 - 2mr^2(E - mgz), \qquad g(r) = 2mr^2(1 + [z']^2). \tag{45}$$

Stability Criterion Derivation

Step 1: First derivative $f'(r)$. Differentiate $f(r) = L_z^2 - 2mr^2(E - mgz(r))$ using the product rule on $r^2 z(r)$:

$$f'(r) = -2m\left[2r(E - mgz) + r^2(-mgz')\right] = -4mr(E - mgz) + 2mgr^2 z'(r). \tag{46}$$

At a circular orbit, $f(r_c) = 0$ from (39). The stationarity condition $U'_{\text{eq}}(r_c) = 0$ is equivalent to $f'(r_c) = 0$ when $f(r_c) = 0$. Setting (46) to zero at $r = r_c$:

$$-4mr_c(E - mgz_c) + 2mgr_c^2 z'_c = 0 \implies E - mgz_c = \frac{mg}{2} r_c z'_c, \tag{47}$$

which reproduces (43).

Step 2: Second derivative $f''(r)$. Differentiate (46) term by term:

$$\frac{d}{dr}[-4mr(E - mgz)] = -4m(E - mgz) - 4mr(-mgz'), \tag{48}$$

$$\frac{d}{dr}[2mgr^2 z'] = 4mgrz' + 2mgr^2 z''. \tag{49}$$

Summing,

$$f''(r) = -4m(E - mgz) + 8mgrz' + 2mgr^2 z''. \tag{50}$$

At $r = r_c$, substitute $E - mgz_c = \frac{mg}{2} r_c z'_c$:

$$\begin{aligned} f''(r_c) &= -4m\left(\frac{mg}{2} r_c z'_c\right) + 8mgr_c z'_c + 2mgr_c^2 z''_c \\ &= -2mgr_c z'_c + 8mgr_c z'_c + 2mgr_c^2 z''_c \\ &= 6mgr_c z'_c + 2mgr_c^2 z''_c. \end{aligned} \tag{51}$$

Step 3: Quotient rule for $U''_{\text{eq}}(r_c)$. With $U_{\text{eq}} = f/g$,

$$U''_{\text{eq}} = \frac{f''g - fg''}{g^2}. \tag{52}$$

At r_c, both $f(r_c) = 0$ and $f'(r_c) = 0$, so the $f'g'$ term vanishes and

$$U''_{\text{eq}}(r_c) = \frac{f''(r_c)}{g(r_c)}. \tag{53}$$

Using $g(r_c) = 2mr_c^2(1 + [z'_c]^2)$:

$$U''_{\text{eq}}(r_c) = \frac{6mgr_c z'_c + 2mgr_c^2 z''_c}{2mr_c^2(1 + [z'_c]^2)} = \frac{g\,(3z'_c + r_c z''_c)}{r_c(1 + [z'_c]^2)}. \tag{54}$$

Stability ($U''_{\text{eq}}(r_c) \geq 0$) with $g > 0$ and $r_c > 0$ requires

Circular-Orbit Stability Criterion

$$3z'(r_c) + r_c z''(r_c) \geq 0. \tag{55}$$

We define the *stability indicator*

$$S(r) \equiv 3z'(r) + rz''(r), \tag{56}$$

so that circular-orbit stability at r_c is equivalent to $S(r_c) \geq 0$ when $z'(r_c) > 0$.

Dimensional note on Eq. (56). Because $z(r)$ and r both carry length, the slope $z' = dz/dr$ is dimensionless ($[z'] = 1$) and the curvature $z'' = d^2z/dr^2$ has dimensions m^{-1}. The product rz'' is therefore dimensionless, and $S(r)$ is a pure number.

Physical Interpretation

This criterion answers whether a circular orbit is a *stable* equilibrium of the radial EAPEF or merely a saddle. When $S(r_c) > 0$, small radial displacements increase U_{eq}, producing a restoring tendency and oscillatory radial motion about r_c. When $S(r_c) < 0$, the EAPEF has a local maximum at r_c and linearized perturbations grow—the circular path is unstable. The indicator depends only on the surface shape (z', z'') at r_c, not on L_z or E directly. For the cone, $S = 3 > 0$ everywhere; for the paraboloid at $r_0 = 1.2\,\text{m}$, $S = 9.6 > 0$; for the Gaussian surface, $S < 0$ at $r = 1.0\,\text{m}$ and $S > 0$ at $r = 2.0\,\text{m}$, though $z' < 0$ prevents real circular orbits there.

4 Dimensional Analysis

Every term in the derived equations must carry consistent SI dimensions. With $[m] = \text{kg}$, $[r] = \text{m}$, $[t] = \text{s}$, $[g] = \text{m/s}^2$, and $[z] = \text{m}$ (so $[z'] = 1$ and $[z''] = 1/\text{m}$):

Table 2: Dimensional verification of key quantities.

Quantity	Dimensions	Consistency check
$L_z = mr^2\dot{\phi}$	$\text{kg}\,\text{m}^2\,\text{s}^{-1}$	verified
E, U_{eq}, mgz	$\text{J} = \text{kg}\,\text{m}^2\,\text{s}^{-2}$	verified
$\frac{1}{2}m\dot{r}^2(1 + [z']^2)$	$\text{kg}\,(\text{m/s})^2$	kinetic energy
$L_z^2/(2mr^2)$	$\text{kg}^2\text{m}^4\text{s}^{-2}/(\text{kg}\,\text{m}^2) = \text{J}$	centrifugal term
U_{eq} numerator $L_z^2 - 2mr^2(E - mgz)$	$\text{J} \cdot \text{kg}$	divided by $2mr^2(1 + [z']^2) \Rightarrow \text{J}$
$\ddot{r}$ in Eq. (38)	m/s^2	numerator $\div[1 + (z')^2]$
$S(r) = 3z' + rz''$	dimensionless	$[z'] = 1$, $[z''] = \text{m}^{-1}$, $[rz''] = 1$
$L_{z,c} = \sqrt{mgr_c^3 z'}$	$\text{kg·m}^2/\text{s}$	verified when $z' > 0$

The EAPEF has the same dimensions as energy because it is constructed from E, $L_z^2/(mr^2)$, and $mgz(r)$, all energy-scaled quantities, with the dimensionless metric factor $1+[z']^2$ in the denominator. The stability indicator $S(r) = 3z' + rz''$ is dimensionless: z' is a slope (length over length) and rz'' is the product of a length with inverse length.

5 Numerical Integration and Error Analysis

5.1 From Derived Equations to Simulation

The integration scheme applies Eqs. (38) and (19) directly:

1. **State variables.** The dynamical state is $(r, \dot{r}, \phi)$. For each run, (L_z, E_0) are fixed from the initial state; E_0 serves as the reference for energy monitoring.

2. **Surface evaluation.** At the current r, evaluate analytically

$$z = z(r), \qquad z' = \frac{dz}{dr}, \qquad z'' = \frac{d^2z}{dr^2}.$$

Closed forms are used for each profile rather than numerical differentiation.

3. **Radial acceleration.** With $m = g = 1$, Eq. (38) becomes

$$\ddot{r} = \frac{L_z^2/r^3 - z' - z'z''\dot{r}^2}{1 + (z')^2}. \tag{57}$$

4. **Explicit Euler updates.** With fixed time step $\Delta t = 0.005\,\text{s}$:

$$\dot{r}_{n+1} = \dot{r}_n + \ddot{r}_n\,\Delta t, \tag{58}$$
$$r_{n+1} = r_n + \dot{r}_{n+1}\,\Delta t, \tag{59}$$
$$\phi_{n+1} = \phi_n + \frac{L_z}{r_n^2}\,\Delta t. \tag{60}$$

   This first-order scheme is not symplectic and does not conserve energy exactly.

5. **Energy monitoring.** After each step, independently compute

$$E_t = \tfrac{1}{2}(1 + z'^2)\dot{r}^2 + \frac{L_z^2}{2r^2} + z$$

   and $\delta_E = |E_t - E_0|/|E_0| \times 100\%$. Because E_t is not fed back into the integrator, agreement between E_0 and E_t validates the ODE implementation.

6. **Turning-point detection.** Periapsis events are identified when $\dot{r}$ changes sign from non-positive to positive. The radial half-period ΔT_r is the time between consecutive such events; $T_a = 2\pi\Delta T_r/\Delta\phi$ follows from the recorded $\Delta\phi$.

5.2 Truncation Error and Energy Drift

Why explicit Euler violates energy conservation. The exact system is Hamiltonian: E is conserved by the continuous-time flow and Liouville's theorem guarantees phase-space volume preservation. Explicit Euler replaces the true trajectory by a first-order Taylor expansion over each step; it is not symplectic, so it neither preserves E exactly nor the canonical phase-space area. This introduces local truncation error $\mathcal{O}(\Delta t^2)$ per step and global energy error $\mathcal{O}(\Delta t)$ over fixed physical time.

Why the paraboloid shows larger drift than the cone. Table 3 lists the maximum δ_E obtained in each run. At $\Delta t = 0.005\,\text{s}$, the cone reaches $\max \delta_E = 0.108\%$ while the paraboloid reaches 0.440%.

Two effects contribute. The paraboloid orbit spans a larger radial range ($\Delta r \approx 1.75\,\text{m}$ vs. $\Delta r \approx 0.94\,\text{m}$ on the cone), so the particle samples regions where $z''\dot{r}^2$ and the metric factor $1 + 4r^2$ vary more strongly. In addition, the curvature term $-z'z''\dot{r}^2$ in Eq. (38) vanishes on the cone ($z'' = 0$) but remains active on the paraboloid ($z'' = 2$).

Time-step halving. Halving Δt from 0.005 s to 0.0025 s reduces maximum δ_E by approximately 50%, consistent with $\mathcal{O}(\Delta t)$ scaling:

The near-factor-of-two reduction confirms that observed drift is dominated by Euler truncation rather than errors in the ODE implementation. A log–log scan over $\Delta t \in \{0.01, 0.005, 0.0025, 0.001\}\,\text{s}$ yields fitted slopes 0.997 (cone) and 0.990 (paraboloid); see Fig. 5 and Sec. 7.

Table 3: Energy drift vs. time step for bounded runs.

Surface	Δt (s)	max δ_E (%)	Final δ_E (%)	Ratio
Cone	0.005	0.107 507	0.005 038	1.00
Cone	0.0025	0.053 847	0.002 498	0.50
Paraboloid	0.005	0.440 049	0.292 182	1.00
Paraboloid	0.0025	0.221 535	0.148 280	0.50

Table 4: Paper claims vs. project verification.

Paper claim [1]	Project verification	Match?
EAPEF $U_{\text{eq}}(r)$ with metric factor $1+[z']^2$	Derived Eq. (26); $U_{\text{eq}}(r_0)$ for cone, paraboloid, Gaussian	Yes
Turning points at $U_{\text{eq}} = 0$	Cone/paraboloid: $\leq$ 0.043% error; Gaussian: single root $\Rightarrow$ escape	Yes
Stability $3z' + rz'' \geq 0$ for stable circular orbits	$S(r_0) = 3.0$ (cone), 9.6 (paraboloid); Gaussian: $z' < 0$	Yes
$L_z = \sqrt{mgr^3z'}$ for circular orbits	$L_{z,c} = 1.315$ (cone), 2.036 (paraboloid) at $r_0 = 1.2\,\text{m}$	Yes
Radial motion from conserved E and L_z	Bounded: $\delta_E \leq 0.44\%$; Gaussian: escape confirmed	Yes
EAPEF superior to U_{eff} for curved surfaces	$U_{\text{eq}} \neq U_{\text{eff}}$ at same r_0 (cone: -0.125 vs. $-0.25\,\text{J}$)	Yes

5.3 Comparison with Timberlake and Mbenoun (2026)

Table 4 compares the principal claims of Ref. [1] with our numerical results.

Every claim listed above is supported by the analytic derivation in Sec. 3 and the numerical results in Sec. 7.

6 Quantitative Analysis of Dynamical Quantities

This section defines the quantities plotted and tabulated in Sec. 7. Each subsection states the formula with $m = g = 1$ and the procedure used to obtain the numerical values.

Table 5 lists each quantity, its defining equation in this section, and the figure or table in Sec. 7 where it appears.

Subsections 6.1–6.7 follow the order of Table 5. Analytic formulas use closed-form $z(r)$ and its derivatives; simulation values are taken directly from the integration output without post-hoc fitting.

6.1 Effective Analogous Potential $U_{\text{eq}}(r)$

For fixed initial energy E_0 and angular momentum L_z, the EAPEF profile is evaluated from Eq. (26) with $m = g = 1$:

$$U_{\text{eq}}(r) = \frac{L_z^2 - 2r^2\big(E_0 - z(r)\big)}{2r^2\big(1 + [z'(r)]^2\big)}. \tag{61}$$

Table 5: Quantities defined in Sec. 6 and where they are reported in Sec. 7.

Quantity	Defining equation(s)	Reported in
$U_{\mathrm{eq}}(r)$, $U_{\mathrm{eff}}(r)$	Eqs. (61)–(62)	Fig. 1; turning-point count
$S(r)$	Eq. (63)	Table 6; Sec. 7.3
$L_{z,c}$	Eq. (64)	Table 6
Turning radii	Eqs. (65)–(66)	Table 6
δ_E	Eqs. (67)–(68)	Figs. 3, 5; Tables 6, 3
ΔT_r, T_a/T_r, $\langle\Delta\phi\rangle/(2\pi)$	Eqs. (69)–(71)	Table 6; Fig. 6
$L_{z,\mathrm{num}}$	Eq. (72)	Sec. 7.2.2
Escape time	t when $r > 10\,\mathrm{m}$	Table 7

For comparison with the uncorrected effective potential, we also plot

$$U_{\mathrm{eff}}(r) = \frac{L_z^2}{2r^2} + z(r) - E_0. \tag{62}$$

On a uniform radial grid $0.05 \le r \le r_{\max}$ with $\Delta r = 0.0005\,\mathrm{m}$ ($r_{\max} = 3.5\,\mathrm{m}$ for cone/paraboloid, $4.0\,\mathrm{m}$ for Gaussian), Eqs. (61)–(62) are evaluated using closed-form $z(r)$, $z'(r)$, and $z''(r)$. A turning radius r_t is recorded when $U_{\mathrm{eq}}(r)$ changes sign across consecutive grid points.

6.2 Stability Indicator $S(r)$

The dimensionless stability indicator from Eq. (56) is evaluated analytically on the same r grid:

$$S(r) = 3z'(r) + r\,z''(r). \tag{63}$$

At a candidate circular radius r_c, local stability of the radial EAPEF requires $S(r_c) \ge 0$ when $z'(r_c) > 0$ (Eq. (55)). This quantity appears in Table 6 at $r = r_0$.

6.3 Circular-Orbit Angular Momentum

When $z'(r_c) > 0$, the angular momentum compatible with a circular orbit at radius r_c is, from Eq. (44) with $m = g = 1$,

$$L_{z,c}(r_c) = \sqrt{r_c^3\, z'(r_c)}. \tag{64}$$

Values at $r_0 = 1.2\,\mathrm{m}$ are compared with the simulated L_z (paraboloid: $L_z = 5.0 > L_{z,c} = 2.036$).

6.4 Turning Points

Analytic prediction. Turning radii satisfy the radial turning-point condition inherited from Eq. (27):

$$U_{\mathrm{eq}}(r_t) = 0 \quad \Longleftrightarrow \quad L_z^2 - 2r_t^2\big(E_0 - z(r_t)\big) = 0. \tag{65}$$

Two distinct roots $r_{t,\mathrm{in}} < r_{t,\mathrm{out}}$ indicate bounded radial motion; a single root indicates outward escape on the Gaussian surface.

Simulation observation. From the full $r(t)$ trajectory, we record $r_{\min}$ and $r_{\max}$ over the integration window. Periapsis events (inner turning points) are detected when $\dot{r}$ changes from non-positive to positive. Relative error is

$$\varepsilon_r = \frac{|r_{\text{obs}} - r_{\text{pred}}|}{r_{\text{pred}}} \times 100\%. \tag{66}$$

Reported cone/paraboloid agreements are $\leq 0.043\%$ (Table 6).

6.5 Energy Monitoring

After each integration step, total energy is *recomputed* from the reduced form of Eq. (21) with $m = g = 1$:

$$E_t = \frac{1}{2}\left(1 + [z'(r)]^2\right)\dot{r}^2 + \frac{L_z^2}{2r^2} + z(r), \tag{67}$$

using the simulated $(r, \dot{r})$ and analytic $z'(r)$. The reference value E_0 is fixed from the initial state. Relative drift is

$$\delta_E(t) = \frac{|E_t - E_0|}{|E_0|} \times 100\%. \tag{68}$$

Because explicit Euler does not enforce $E_t = E_0$, $\max \delta_E$ over the run quantifies integrator truncation error (Sec. 7.2.1, Table 3). E_t is never fed back into the integrator.

6.6 Radial and Azimuthal Periods

Radial half-period. Between successive periapsis events at times t_{k-1} and t_k,

$$\Delta T_r = t_k - t_{k-1}. \tag{69}$$

Azimuthal advance and effective azimuthal period. Over the same interval, the azimuthal advance is $\Delta\phi_k = \phi(t_k) - \phi(t_{k-1})$. The corresponding azimuthal period estimate is

$$T_a = \frac{2\pi\,\Delta T_r}{\Delta\phi}, \qquad \frac{T_a}{\Delta T_r} = \frac{2\pi}{\Delta\phi}. \tag{70}$$

The mean $\langle\Delta\phi\rangle$ per radial cycle and the ratio $\langle\Delta\phi\rangle/(2\pi)$ are recorded at each periapsis. Measured values 0.804 (cone) and 1.864 (paraboloid) are not integer multiples of 2π, so the horizontal projection does not close (Fig. 6). The closure residual

$$\left| \frac{\Delta\phi}{2\pi} - \text{round}\left(\frac{\Delta\phi}{2\pi}\right) \right| \tag{71}$$

quantifies how far each radial cycle falls from closing in the horizontal plane.

6.7 Angular Momentum Check

The reduced equations hold L_z fixed. As a numerical consistency check, each step records

$$L_{z,\text{num}} = r^2\,\frac{\Delta\phi}{\Delta t} \tag{72}$$

from the ϕ increment at each step and compares it with the nominal L_z. Relative deviations remain below $10^{-9}\%$ on bounded runs, so the observed energy drift is due to Euler truncation rather than loss of L_z.

Sec. 7 applies these definitions: Sec. 7.2 reports the cone and paraboloid results in Table 6; Sec. 7.3 treats the Gaussian escape runs in Table 7.

7 Results

Sec. 7.2 presents detailed results for the cone and paraboloid: turning-point errors, energy drift, period ratios, and azimuthal precession (Fig. 6). Sec. 7.3 treats the Gaussian surface $z = e^{-r^2}$, for which the EAPEF has a single zero and the particle escapes outward. Figs. 1–4 compare all three surfaces side by side.

7.1 Simulation Figures

The figures below were generated from the Euler integration in Sec. 5.1. Each time series was subsampled to at most 320 points per curve for plotting.

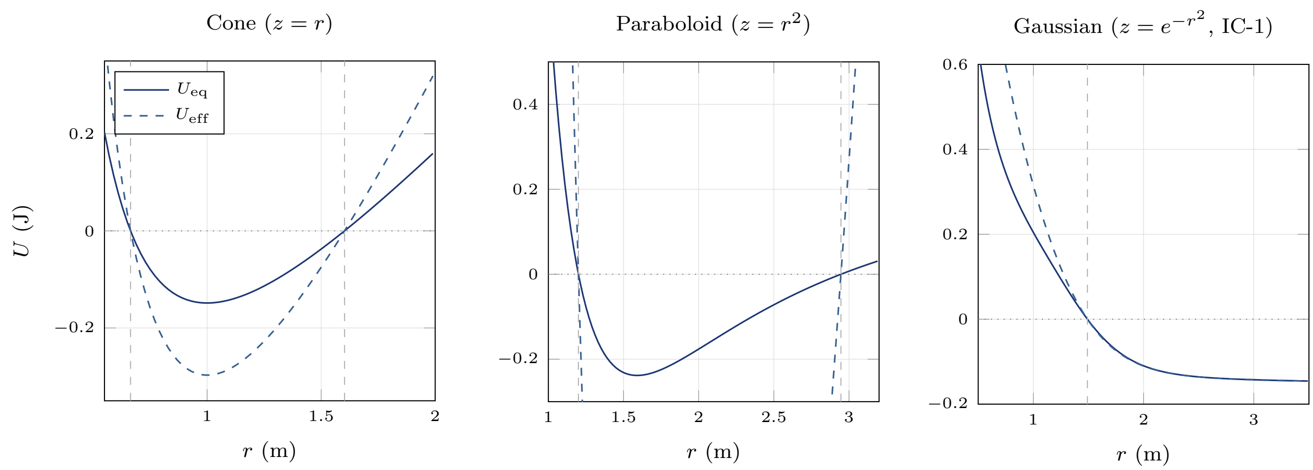

Figure 1: Effective analogous potential $U_{eq}(r)$ and uncorrected $U_{eff}(r)$ at fixed (E_0, L_z). Cone and paraboloid (left/center): two $U_{eq} = 0$ roots bracket a radial oscillation band. Gaussian (right): a single root and outward escape.

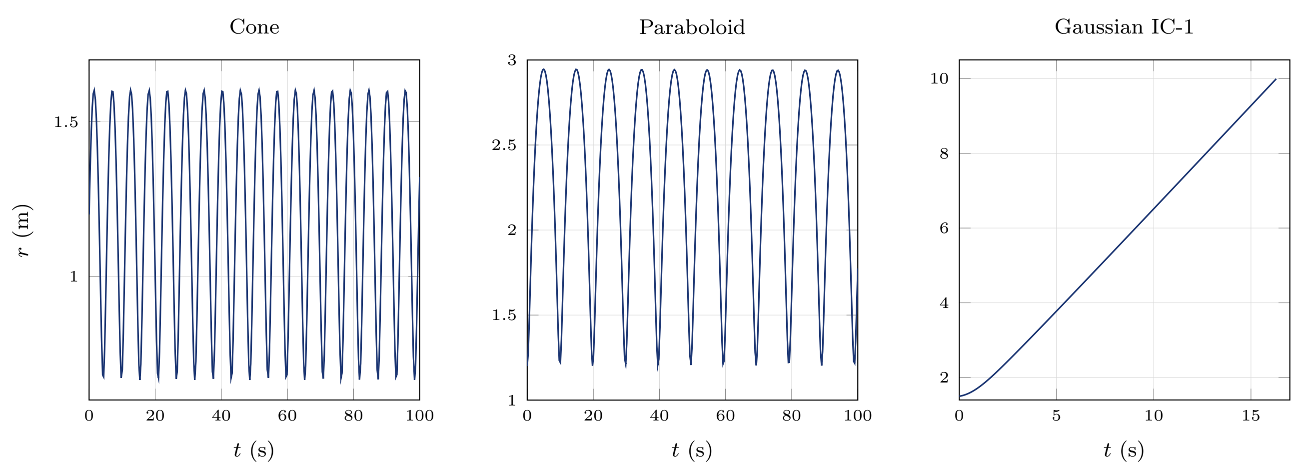

Figure 2: Radial coordinate $r(t)$ ($\Delta t = 0.005\,$s). Cone and paraboloid: periodic oscillation between turning radii. Gaussian IC-1: monotonic escape at $t = 16.315\,$s.

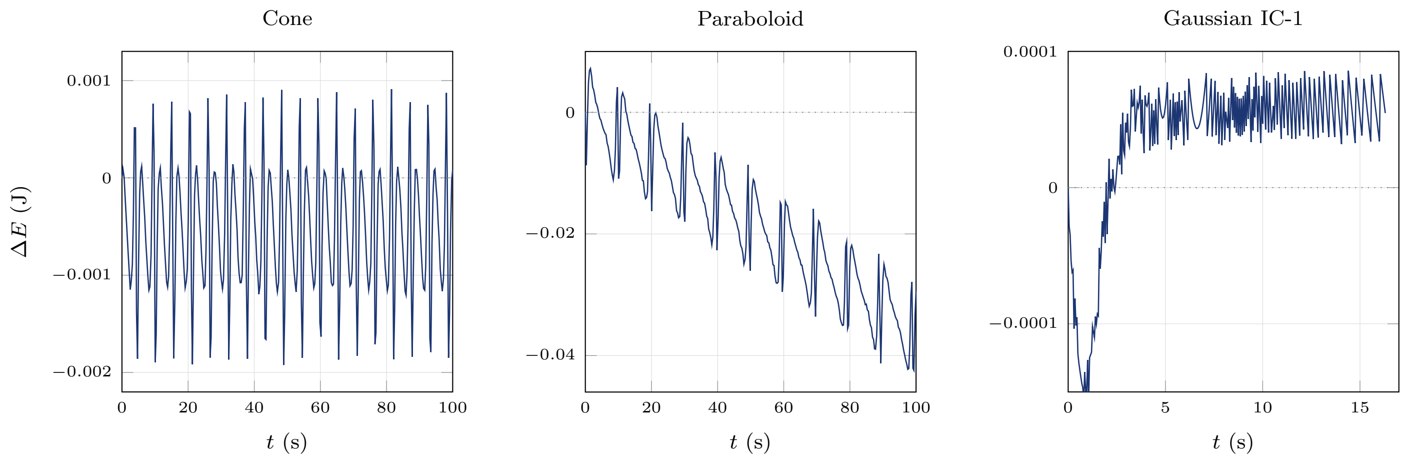

Figure 3: Energy deviation $\Delta E = E(t) - E(0)$. Cone and paraboloid: $\max \delta_E = 0.11\%$ and 0.44%. The Gaussian panel uses a magnified vertical scale; escape trajectories are not characterized by periodic energy oscillation.

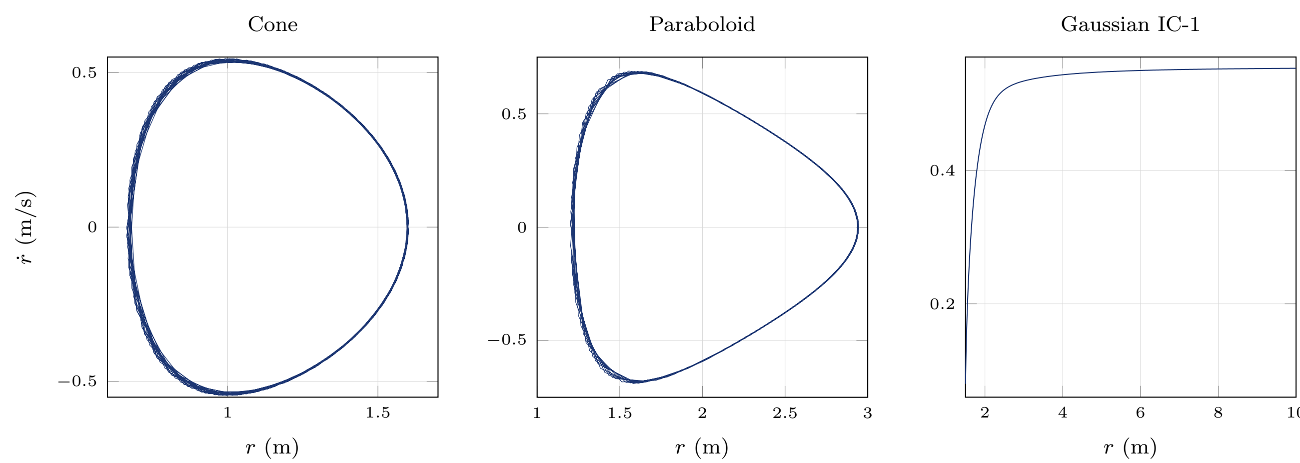

Figure 4: Phase-space trajectories $(r, \dot{r})$. Cone and paraboloid: closed loops between turning radii. Gaussian IC-1: monotonic escape branch.

7.2 Bounded Motion

The cone $z = r$ and paraboloid $z = r^2$ each have two zeros of $U_{\text{eq}}(r)$, confining the particle between inner and outer turning radii. Table 6 summarizes the initial conditions, turning-point agreement, periodicity, and energy drift ($m = g = 1$, $\Delta t = 0.005$ s).

Table 6: Bounded-motion results: cone and paraboloid ($m = g = 1$, $\Delta t = 0.005$ s).

Quantity	Cone ($z = r$)	Paraboloid ($z = r^2$)
Initial conditions		
r_0 (m)	1.2	1.2
$\dot{r}_0$ (m/s)	0.5	0
L_z (kg m^2/s)	1.0	5.0
t_{max} (s)	100	100
Energy and EAPEF at $t = 0$		
E_0 (J)	1.7972	10.1206
$\max \delta_E$ (%)	0.1075	0.4400
Final δ_E (%)	0.0050	0.2922
$U_{\text{eq}}(r_0)$ (J)	-0.1250	0
$U_{\text{eff}}(r_0)$ (J)	-0.2500	0
$S(r_0)$	3.0	9.6
$L_{z,c}(r_0)$ (kg m^2/s)	1.3145	2.0365
Turning radii: prediction vs. simulation		
Inner r_t pred. (m)	0.6643	1.2000
Inner r_t obs. (m)	0.6646	1.2000
Inner rel. error (%)	0.0326	0
Outer r_t pred. (m)	1.6025	2.9463
Outer r_t obs. (m)	1.6018	2.9459
Outer rel. error (%)	0.0433	0.0122
Periodic radial dynamics		
r_{min} (m)	0.6646	1.2000
r_{max} (m)	1.6018	2.9459
Periapsis count	18	11
ΔT_r (s)	5.540	9.885
$\langle \Delta\phi \rangle/(2\pi)$	0.8043	1.8644
T_a/T_r	1.244	0.5364

7.2.1 Energy Drift and Time-Step Scaling

Explicit Euler introduces $\mathcal{O}(\Delta t)$ global energy error on bounded orbits (Sec. 5.2, Table 3). At $\Delta t = 0.005$ s, the paraboloid's wider radial excursion ($\Delta r \approx 1.75$ m) and active curvature term $z'z''\dot{r}^2$ in Eq. (38) yield $\max \delta_E = 0.440\%$, four times the cone value (0.108%). Halving Δt halves $\max \delta_E$ to good approximation.

Using the same initial conditions, integrations to $t = 100$ s were repeated at $\Delta t \in \{0.01, 0.005, 0.0025, 0.001\}$ s. Fig. 5 plots $\log_{10}(\max \delta_E)$ versus $\log_{10}(\Delta t)$; least-squares slopes are 0.997 (cone) and 0.990 (paraboloid).

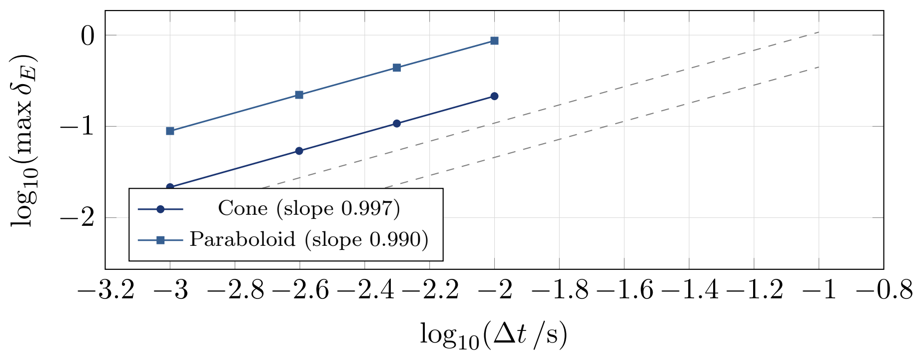

Figure 5: Log–log plot of maximum energy drift vs. time step for bounded runs. Slopes near unity confirm first-order Euler scaling (Table 3).

7.2.2 Azimuthal Precession

Between consecutive periapsis events, the simulation records $\Delta\phi$ and ΔT_r. Mean values $\langle\Delta\phi\rangle/(2\pi) = 0.804$ (cone) and 1.864 (paraboloid) show that each radial cycle advances the azimuth by a non-integer multiple of 2π, so the horizontal projection precesses ($T_a/T_r = 1.244$ and 0.536; Fig. 6). Angular momentum is preserved to better than $10^{-9}\%$, so the energy drift is due to the integrator rather than loss of L_z.

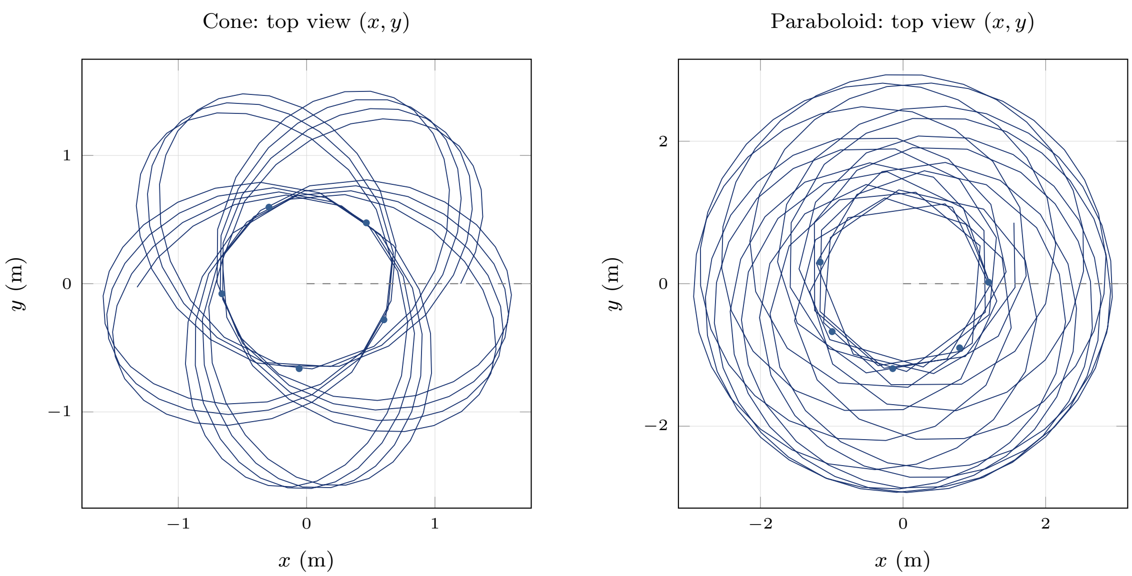

Figure 6: Horizontal-plane projection $(x, y) = (r\cos\phi, r\sin\phi)$ over $t = 100\,\mathrm{s}$. Blue curves: trajectory; orange markers: periapsis events. Successive radial cycles do not overlap in angle ($\langle\Delta\phi\rangle/(2\pi) = 0.804$ cone, 1.864 paraboloid).

Physical Interpretation

- **Cone** (left panels of Figs. 1–4): $U_{eq} < 0$ between $r = 0.6643$ m and $r = 1.6025$ m. Turning radii match $U_{eq} = 0$ to $\leq 0.043\%$; $\max \delta_E = 0.108\%$. The $(r, \dot{r})$ portrait is a closed loop, and the top view in Fig. 6 precesses with $\langle \Delta\phi \rangle/(2\pi) = 0.804$ per radial cycle.

- **Paraboloid** (center panels): with $L_z = 5.0 > L_{z,c} = 2.036$, the orbit spans $r \in [1.20, 2.95]$ m. Outer turning radius agrees to 0.012%; $\max \delta_E = 0.44\%$ is the largest among bounded runs. The wider (x, y) trace in Fig. 6 reflects stronger azimuthal advance ($\langle \Delta\phi \rangle/(2\pi) = 1.864$).

7.3 Unbounded Motion on the Gaussian Surface

For the Gaussian profile $z(r) = e^{-r^2}$, the surface slopes downward for $r > 0$ ($z' < 0$), so $L_{z,c} = \sqrt{r^3 z'(r)}$ is not real. A parameter scan over 28 665 initial conditions finds no case with two U_{eq} zeros; every trial has exactly one root. Both escape runs in Table 7 show monotonic outward growth to $r > 10$ m with no periapsis recurrence.

Table 7: Gaussian escape runs: single-root EAPEF and outward motion ($m = g = 1$, $\Delta t = 0.005$ s).

Quantity	IC-1	IC-2
r_0 (m)	1.5	1.0
$\dot{r}_0$ (m/s)	0.08	0
L_z (kg m^2/s)	0.45	0.25
t_{max} (s)	20	15
E_0 (J)	0.1539	0.3991
$U_{eq}(r_0)$ (J)	−0.0032	0
U_{eq} zero (m)	1.4907	1.0000
$S(r_0)$	0.1581	−1.4715
Escape time (s)	16.315	10.985
r_{max} reached (m)	10.0013	10.0021
Bounded? (2 roots)	No (1 root each)	

Physical Interpretation

On a downhill surface with $z' < 0$, the EAPEF admits only one turning radius. IC-1 crosses $U_{eq} = 0$ at $r = 1.4907$ m and escapes at $t = 16.3$ s; IC-2 starts on that root ($U_{eq}(r_0) = 0$, $\dot{r}_0 = 0$) yet immediately departs because circular force balance is impossible when gravity pulls along a descending tangent. Figs. 1–4 (right panels) show the single EAPEF zero and the monotonic escape branch.

8 Discussion

Our results fall into two groups: quantitative comparison for bounded cone and paraboloid orbits, and a qualitative check of escape on the Gaussian surface. The same integration scheme is used throughout; only the surface profile $z(r)$ changes.

Cone and paraboloid. On the cone ($z' = 1$, $z'' = 0$), the metric factor $1 + [z']^2 = 2$ is constant and the curvature term in Eq. (38) vanishes. Turning radii 0.6643 m and 1.6025 m bracket a narrow band ($\Delta r \approx 0.94$ m) with max $\delta_E = 0.108\%$ and a precessing horizontal projection ($T_a/T_r = 1.244$; Fig. 6, left). On the paraboloid, the larger radial excursion ($\Delta r \approx 1.75$ m) and active $z'z''\dot{r}^2$ coupling raise max δ_E to 0.440% at the same Δt, yet turning points still agree to 0.012%. Log–log slopes 0.997 and 0.990 (Fig. 5) indicate that the larger drift on the paraboloid reflects orbit amplitude and Euler truncation, not a failure of the EAPEF ODE.

Gaussian escape. The Gaussian bell was included to test whether a surface with $z' < 0$ and a single U_{eq} zero produces outward escape, as the analytic framework predicts. Both runs in Table 7 reach $r > 10$ m within 11–16 s with no periapsis recurrence, consistent with the absence of an outer turning wall.

Summary. The EAPEF correctly predicts turning radii at the 0.04% level on bounded orbits and identifies escape when only one root exists. Energy drift on bounded runs scales as $\mathcal{O}(\Delta t)$ and remains below 0.5% at $\Delta t = 0.005$ s. Because explicit Euler is not symplectic, $(r, \dot{r})$ phase areas can distort even while L_z is preserved to round-off precision.

9 Conclusion

This study derived and numerically checked the EAPEF framework for motion on a surface of revolution [1].

Derivation. We obtained the reduced Lagrangian, proved L_z conservation, derived $U_{\text{eq}}(r)$ and the radial ODE, and established the circular-orbit condition $L_{z,c} = \sqrt{mgr_c^3 z'(r_c)}$ and stability criterion $S(r) = 3z' + rz'' \geq 0$ (Table 2, Sec. 3.9).

Numerical results. Explicit Euler integration yielded turning-point agreement at the 0.04% level on the cone and paraboloid, period ratios $T_a/T_r = 1.244$ and 0.536, azimuthal precession $\langle \Delta\phi \rangle/(2\pi) = 0.804$ and 1.864 per radial cycle (Fig. 6), and energy drifts below 0.44% over 100 s (Table 6, Fig. 5).

Gaussian escape. For $z = e^{-r^2}$, the EAPEF has a single zero and both initial conditions in Table 7 lead to outward escape, as predicted analytically.

Comparison with Ref. [1]. Table 4 shows agreement with the principal claims tested here. On the cone, $U_{\text{eq}} \neq U_{\text{eff}}$ at the same initial state illustrates the improvement over the uncorrected effective potential on curved surfaces.

Limitations. Explicit Euler introduces $\mathcal{O}(\Delta t)$ energy drift on bounded orbits (Table 3, Fig. 5) and is not symplectic. Paraboloid orbits with large radial amplitude show the largest absolute $|E(t) - E(0)|$ among the bounded runs.

Extensions. Higher-order or symplectic integrators could reduce energy drift while keeping the same ODE. Other profiles such as $z(r) = r^3$ or piecewise surfaces would further test the stability indicator; friction or time-dependent surfaces $z(r, t)$ lie outside the present conservative framework.

Within the tested parameter ranges, the agreement between U_{eq} predictions and simulated orbits supports the EAPEF as a valid Newtonian description of surface-of-revolution dynamics.

References

[1] T. K. Timberlake and R. Mbenoun, “Analyzing motion on a surface of revolution using the effective analogous potential energy function,” *Am. J. Phys.* **94**, 16 (2026).

[2] J. B. Marion and S. T. Thornton, *Classical Dynamics of Particles and Systems*, 5th ed. (Brooks/Cole, 2003).